

JEE MAIN REVISION MASTER: APPLICATION OF DERIVATIVES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 33 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Tangents, Normals & Rate of Change	18 PYQs	Tangents & Rates
Very High	Monotonicity, Maxima & Minima	15 PYQs	Optimization

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Tangents, Normals & Rate of Change	$m = \left(\frac{dy}{dx}\right)_{(x_1, y_1)}$, $\text{Tangent: } y - y_1 = m(x - x_1)$, $\text{Normal: } y - y_1 = -\frac{1}{m}(x - x_1)$
Monotonicity, Maxima & Minima	$f'(x) > 0 \implies \text{Strictly Increasing}$, $f'(c) = 0$ & $f''(c) < 0 \implies \text{Local Max}$, $f''(c) > 0 \implies \text{Local Min}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Tangents, Normals & Rate of Change (Very High Yield — 18 PYQs)

Core Principle: Equations of tangent and normal lines, angle of intersection between two curves, and time derivatives as physical rates.

■ **Caution / JEE Pitfall:** Two curves cut orthogonally if and only if the product of their tangent slopes at the intersection point is -1 ($m_1 m_2 = -1$).

■ **Speed Trick:** Length of sub-tangent = $\left|y / \frac{dy}{dx}\right|$; length of sub-normal = $|y \cdot (dy / dx)|$.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "If $f(x) = xe^x(1-x)$, then $f(x)$ is..." [Ans: N/A]
- [2024 Shift] Q#4 (MCQ): "The length of a rectangle is five times the breadth. If the minimum perimeter of the rectangle is 180 cm, then..." [Ans: N/A]
- [2024 Shift] Q#4 (MCQ): "The maximum volume of the right circular cone with slant height 6 units is..." [Ans: N/A]

3.2 Monotonicity, Maxima & Minima (Very High Yield — 15 PYQs)

Core Principle: First and second derivative tests for optimization, global extreme values on closed intervals, and Rolle's/Lagrange's Mean Value Theorems.

■ **Caution / JEE Pitfall:** A critical point occurs where $f'(x) = 0$ OR where $f'(x)$ DOES NOT EXIST (e.g. at $x=0$ for $|x|$).

■ **Speed Trick:** LMVT statement: If f is continuous on $[a, b]$ and differentiable on (a, b) , then $\exists c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A YouTube short video is getting viral according to $f(t) = -2t^3 + 3t^2 + 5$. At what time does the video get maximum number of shares? (t is in hours)" [Ans: N/A]
- **[2026 Shift] Q#2 (MCQ):** "In a Mahakumbh, a drone camera is moving along $3y = x^3 - 3$. When y -coordinate changes 9 times as fast as x -coordinate, it captures..." [Ans: C]
- **[2025 Shift] Q#3 (MCQ):** "The function $f(x) = \tan x - x$..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A YouTube short video is getting viral according to $f(t) = -2t^3 + 3t^2 + 5$. At what time does the video get maximum number of shares? (t is in hours)

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Correct Answer: None

Solution Analysis: The number of shares is given by $f(t) = -2t^3 + 3t^2 + 5$. To find when the video gets the maximum number of shares, we differentiate $f(t)$ with respect to t . $f'(t) = \frac{d}{dt}(-2t^3 + 3t^2 + 5) = -6t^2 + 6t$. For maximum or minimum value, put $f'(t) = 0$. $-6t^2 + 6t = 0$. $6t(1-t) = 0$. So, $t = 0$ or $t = 1$. Now check the nature using the second derivative. $f''(t) = -12t + 6$. At $t = 1$, $f''(1) = -12(1) + 6 = -6$. Since this is negative, $f(t)$ has a maximum at $t = 1$. So, the video gets the maximum number of shares after 1 hour. Correct option: Option A.

Benchmark Problem 2: KCET 2024 (Q#1)

If $f(x) = xe^x(1-x)$, then $f(x)$ is

- (A) increasing in $\mathbb{R}$
- (B) decreasing in $\mathbb{R}$
- (C) decreasing in $[-1/2, 1]$
- (D) increasing in $[-1/2, 1]$

Correct Answer: None

Solution Analysis: Given, $f(x) = xe^x(1-x)$. $f'(x) = e^x(1-x) + xe^x(1-x)(-1) = e^x(1-x)(1-x-1) = -e^x(1-x)(x)$. So, $f(x)$ is increasing when $f'(x) \geq 0$ and decreasing when $f'(x) \leq 0$. Hence, f is increasing in $[-1/2, 1]$.